

Numerical investigation on motion characteristic of a bubble rising along overheated surfaces with different curvatures

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ABSTRACT

In present work, a two-phase thermal LBM model for simulating liquid-gas phase transition in body-fitted coordinate system is proposed. Subsequently, model validation is performed by simulating bubble growth in the superheated liquid without consideration of gravity. After that, the hydrodynamic and thermodynamic behaviors of a bubble during its rising and sliding process along overheated surfaces are analyzed and compared in three cases with different surface axial ratios. The results indicate that bubble experiences the largest deformation and growth in the case with surface axial ratio $a/b = 1.5$. As for bubble average velocity, its horizontal component increases with surface axial ratio, yet opposite for vertical one. The influences of bubble deformation and growth on temperature field become more significant in the case with a larger axial ratio. Besides, the existence of bubble causes an extremely high temperature gradient on the surface during its sliding stage; such effect is weakened gradually as the bubble departs from surface and moves to the far field.

1. Introduction

As a frequently encountered phenomenon in nature and engineering fields, bubble sliding and rising along surface with various geometries has a significant impact on heat and mass transfer [1]. For instance, in the heat exchangers, bubbles may generate on the heated tube surface. The motion of bubble and phase transition between bubble and its surrounding liquid may lead to intense disturbance of temperature and velocity fields, which affects the efficiency of heat transfer significantly [2]. Hence, for the sake of guiding device operation and design, it is necessary to have a further insight of liquid-gas phase transition occurring on surface with complex geometries [3].

Actually, the hydrodynamic and thermodynamic behaviors of a bubble during its interaction process with surfaces have generated considerable recent research interest. With the assistance of high-speed digital cameras, bubble motions were experimentally investigated over an inclined surface [4,5] and a curved tube surface [6]. However, experimental techniques still have difficulty in quantitative prediction due to the complexity of multiphase flows. Fortunately, the rapid advances in computer technology and numerical simulation methods provide a powerful tool to predict bubble behavior. For example, Yan and Li [7] simulated a vapor bubble growing and rising in uniformly superheated liquid by time dependent mathematical and numerical models proposed in their previous works [8,9]. Son et al. [10]

investigated a single bubble during partial nucleate boiling on a horizontal surface by the level set method.

As an alternative numerical approach, the lattice Boltzmann method (LBM) based on kinetic theory has been developed to be a remarkable numerical tool. Due to its distinct advantages such as explicit scheme, parallel computing, and simple boundary treatment, LBM is widely applied for modeling multiphase fluid flow [11–13]. Currently, a considerable number of multiphase LBM models are available, including the chromodynamics model proposed by Rothman and Keller [14], the pseudo-potential method of Shan and Chen [15], the free energy method of Swift et al. [16], and the method of He et al. [17]. Based on these models, Hazi et al. [18] studied heterogeneous boiling on a horizontal plate in stagnant and slowly flowing fluid by adding a phase change equation to LBM two-phase model; Gong et al. [19] proposed an improved thermal LBM model by considering energy equation to simulate phase change between liquid and gas. In addition, by using the single-component multiphase pseudopotential (SCMP) LB method, Gong et al. [20] studied the thermal immiscible multiphase flow. Their work gives a significant inspiration for solving the unphysical mass transfer problem of the SCMP LB model in multi-bubble/droplet systems. Notwithstanding the promise the above-mentioned models showing in multiphase flow simulation, most of these models are limited to small density ratio. To overcome such limitation, Inamuro et al. [21] proposed a lattice Boltzmann method for incompressible two-

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phase flows with large density differences on the basis of the free-energy model. Afterwards, Zheng et al. [22] developed a multiphase model (ZSC model) for the two-phase system with large density ratio by using the Cahn-Hilliard (C-H) equation to track and define two-phase interface, which is closer to the Landau mean-field theory than other models. For the purpose of considering phase transition, Dong et al. [23] proposed a hybrid thermal LBM model by combining ZSC model with thermal LBM model of Inamuro et al. [24], and then applied it to study bubble growth on and departure from a superheated wall.

Despite many successful applications of the LBM in multiphase flow researches, the requirement of uniform grids in standard LBM model still restricts its broad application. In most of practical situations, bubbles move in domain with irregular boundaries, thus the nodes of uniform grids cannot be directly located on the boundaries. In order to accurately describe the research domain, sufficient uniform grids with a stair-shaped approximation for irregular boundaries are needed, which unavoidably deteriorates computational efficiency. Apart from that, accurate curved boundary treatments, including adaptive meshes refinement method and inter/extrapolated curved boundary method, were reported to deal with curved boundaries [25–28]. These boundary schemes, with different numerical stabilities, have their own validities. Alternatively, it is a pretty good way to choose non-uniform grids, which can describe curved boundaries more accurately. Meanwhile, great efforts have been made to develop LBM schemes on non-uniform grids. What will be emphasized is the interpolation-supplemented lattice Boltzmann method (ISLBM) proposed by He and Doolen [29]. As the grids are non-uniform, the particles generally cannot arrive to their neighboring nodes over a time step. Consequently, this method consists of three steps: collision, streaming, and interpolation. After the streaming step, particle distribution functions on lattice nodes are determined by an interpolation scheme. Nevertheless, the requirement of an analytical function to link the transformation between the physical plane and the computational plane hampers its application. By introducing the body-fitted coordinates into the LBM, Imamura et al. [30,31] broke the limitation of ISLBM and developed the generalized form of interpolation-supplemented LBM (GILBM). It was successfully applied to simulate 2D flow around a circular cylinder and the NACA0012 airfoil, respectively. A recent paper by Li et al. [3] proposed a two-phase LBM model in body-fitted coordinates with large density ratio, and it was adopted to study a single bubble sliding and departing from a curved surface in isothermal system.

Obviously, the GILBM-related models have been used to solve single/multi-phase fluid flow in configurations with irregular boundaries. However, to our best knowledge, the liquid-gas phase transition does not be involved in these models so far. The purpose of present work is to develop a two-phase thermal LBM model for simulating liquid-gas phase transition in body-fitted coordinates. Then, it is applied to study the motion characteristic of a single bubble rising along overheated surfaces with different curvatures, as well as the impact of surface configuration on heat and mass transfer. The rest of the article is organized as follows. In Section 2, the two-phase thermal LBM model for simulating phase transition in body-fitted coordinates is presented in detail. Subsequently, the proposed model is validated by simulating the growth of bubble in the overheated liquid without considering gravity in Section 3. In Section 4, numerical simulations and results are presented and discussed. Finally, some conclusions are summarized in Section 5.

2. Methodology

2.1. Hybrid thermal LBM model for phase transition in Cartesian coordinate system

The hybrid thermal LBM model for phase transition in Cartesian coordinate system of Dong et al. [23] is established on the basis of the ZSC model and the thermal LBM model of Inamuro et al. Firstly, order

parameter ϕ and mean density n are defined as

$$\phi = \frac{\rho_l - \rho_g}{2}, \quad n = \frac{\rho_l + \rho_g}{2} \quad (1)$$

where ρ_l and ρ_g are densities of liquid and gas, respectively. For recovering the interface capturing equation (C-H equation), the continuity and momentum equations, and the energy equation, three sets of particle distribution functions (g_i , f_i , and h_i) are introduced, respectively. Their evolution equations are expressed as

$$g_i(\mathbf{x} + \mathbf{e}_i \delta t, t + \delta t) = g_i(\mathbf{x}, t) - \frac{1}{\tau_g} [g_i(\mathbf{x}, t) - g_i^{eq}(\mathbf{x}, t)] + \omega_i \dot{\phi} \quad (2)$$

$$f_i(\mathbf{x} + \mathbf{e}_i \delta t, t + \delta t) = f_i(\mathbf{x}, t) - \frac{1}{\tau_f} [f_i(\mathbf{x}, t) - f_i^{eq}(\mathbf{x}, t)] + \left(1 - \frac{1}{2\tau_f}\right) \frac{3\omega_i}{c^2} [(e_i \cdot \mathbf{u}) + \frac{3(e_i \cdot \mathbf{u})}{c^2} e_i] (\mu_\phi \nabla \phi + \mathbf{F}) \delta t \quad (3)$$

$$h_i(\mathbf{x} + \mathbf{e}_i \delta t, t + \delta t) = h_i(\mathbf{x}, t) - \frac{1}{\tau_h} [h_i(\mathbf{x}, t) - h_i^{eq}(\mathbf{x}, t)] + \omega_i \frac{\rho_g \dot{\phi}}{\rho_l(\rho_l - \rho_g)Ja} \quad (4)$$

where $\mathbf{x}$ is lattice position, $\mathbf{e}_i$ is the discrete particle velocity of i th direction, t and δt are time and time step, respectively. τ_g , τ_f , and τ_h are dimensionless relaxation times. $\mathbf{F}$ is body force, ω_i is weighting coefficient, c is lattice speed, $\mathbf{u}$ is macroscopic velocity. μ_ϕ is chemical potential with the form of

$$\mu_\phi = A(4\phi^3 - 4\phi^{*2}\phi) - \kappa \nabla^2 \phi \quad (5)$$

where ϕ^* is the equilibrium state of order parameter ϕ . And ϕ^* is determined by initial fluid densities. A and κ are coefficients related to the surface tension coefficient σ and the interface layer thickness W as

$$\kappa = \frac{3\sigma W}{8\phi^{*2}} \quad (6)$$

$$A = \frac{3\sigma}{4\phi^{*4}W} \quad (7)$$

For the two-dimensional nine-velocity (D2Q9) LBM model, the discrete velocities $\mathbf{e}_i$ and weighting coefficients ω_i are given as

$$\mathbf{e}_i = \begin{bmatrix} 0 & 1 & 0 & -1 & 0 & 1 & -1 & -1 & 1 \\ 0 & 0 & 1 & 0 & -1 & 1 & 1 & -1 & -1 \end{bmatrix} \quad (8)$$

$$\omega_i = \left[\frac{4}{9}, \frac{1}{9}, \frac{1}{9}, \frac{1}{9}, \frac{1}{9}, \frac{1}{36}, \frac{1}{36}, \frac{1}{36}, \frac{1}{36} \right] \quad (9)$$

Then, the corresponding equilibrium distribution functions g_i^{eq} , f_i^{eq} , and h_i^{eq} can be calculated by

$$g_i^{eq} = \omega_i \left(B_i + \phi \frac{3\mathbf{e}_i \cdot \mathbf{u}}{c^2} \right) \quad (10a)$$

$$B_i = \begin{cases} 3M\mu_\phi/c^2, & i > 0 \\ [\phi - (1 - \omega_0)B]/\omega_0, & i = 0 \end{cases} \quad (10b)$$

$$f_i^{eq} = \omega_i A_i + \omega_i n \left[3\mathbf{e}_i \cdot \mathbf{u} - \frac{3}{2}u^2 + \frac{9}{2}(\mathbf{e}_i \cdot \mathbf{u})^2 \right] \quad (11a)$$

$$A_i = \begin{cases} 3(\phi\mu_\phi + n/3), & i > 0 \\ 9n/4 - 15(\phi\mu_\phi + n/3)/4, & i = 0 \end{cases} \quad (11b)$$

$$h_i^{eq} = \omega_i T(1 + 3\mathbf{e}_i \cdot \mathbf{u}) \quad (12)$$

where the parameter M is defined by mobility coefficient Γ and τ_g as $M = \Gamma/(\tau_g - 0.5)$ [32], T is temperature.

According to the Landau mean-field theory [33], phase transition is considered as a continuous change of order parameter ϕ . Thus, a source

term $\dot{\phi}$ is added to the right-hand side of Eq. (2) to consider phase transition effect [23,34], which can be determined by

$$\frac{dV}{dt} = -\frac{\rho_l Ja}{\rho_g Pe} \left(\frac{\partial T^2}{\partial x^2} \right) = \frac{\dot{\phi}}{\rho_l - \rho_g} \quad (13)$$

where V stands for bubble volume. The Jakob number and Peclet number are defined as $Ja = C_{pl}(T'_b - T'_\infty)/h_{fg}$ and $Pe = \rho_l U_T d_e C_{pl}/\lambda_l$, respectively. Here, C_{pl} is specific heat at constant pressure. T'_b and T'_∞ are the temperature of bubble and its surrounding liquid, respectively. h_{fg} is latent heat, λ_l is thermal conductivity. U_T is the terminal velocity of bubble, and d_e is the equivalent diameter of bubble. Meanwhile, the corresponding latent heat term $\rho_g \dot{\phi}/[\rho_l(\rho_l - \rho_g)Ja]$ derived from phase transition is added to Eq. (4). It should be noted that the source terms in Eqs. (2) and (4) are only applied for $\phi < 0$ (the gas phase). As for $\phi > 0$ (the liquid phase), they should be removed.

The macroscopic variables, including order parameter, mean density, velocity, and temperature, are relate to g_i , f_i , and h_i as

$$\phi = \sum_i g_i \quad (14a)$$

$$n = \sum_i f_i \quad (14b)$$

$$\mathbf{u} = \left[\sum_i f_i \cdot \mathbf{e}_i + (\mu_\phi \nabla \phi + \mathbf{F})/2 \right] / n \quad (14c)$$

$$T = \sum_i h_i \quad (14d)$$

2.2. Two-phase thermal LBM model for simulating liquid-gas phase transition in body-fitted coordinate system

In order to develop a two-phase thermal LBM model in body-fitted coordinates, the governing equations, Eqs. (2)–(4), on the physical plane should be transformed into the computational plane. Here, the physical and computational planes are denoted by $\mathbf{x} = (x_1, x_2) = (x, y)$ and $\xi = (\xi_1, \xi_2) = (\xi, \eta)$, respectively. In the collision equations, the transformation is easily achieved by replacing $\mathbf{x}$ with ξ as

$$g_i^*(\xi, t) = g_i(\xi, t) - \frac{1}{\tau_g} [g_i(\xi, t) - g_i^{eq}(\xi, t)] + \omega_i \dot{\phi} \quad (15)$$

$$f_i^*(\xi, t) = f_i(\xi, t) - \frac{1}{\tau_f} [f_i(\xi, t) - f_i^{eq}(\xi, t)] + \left(1 - \frac{1}{2\tau_f} \right) \frac{3\omega_l}{c^2} \left[(\tilde{\mathbf{e}}_i - \mathbf{u}) + \frac{3(\tilde{\mathbf{e}}_i \cdot \mathbf{u})}{c^2} \tilde{\mathbf{e}}_i \right] (\mu_\phi \nabla \phi + \mathbf{F}) \delta t \quad (16)$$

$$h_i^*(\xi, t) = h_i(\xi, t) - \frac{1}{\tau_T} [h_i(\xi, t) - h_i^{eq}(\xi, t)] + \omega_i \frac{\rho_g \dot{\phi}}{\rho_l(\rho_l - \rho_g)Ja} \quad (17)$$

where g_i^* , f_i^* , and h_i^* are post-collision distribution functions. In Eq. (16), $\tilde{\mathbf{e}}_i = (\tilde{e}_{i,1}, \tilde{e}_{i,2})$ is the contravariant velocity on the computational plane, and its components are defined as

$$\tilde{e}_{i,1} = e_{i,1} \frac{\partial \xi_1}{\partial x_1} + e_{i,2} \frac{\partial \xi_1}{\partial x_2} \quad (18a)$$

$$\tilde{e}_{i,2} = e_{i,1} \frac{\partial \xi_2}{\partial x_1} + e_{i,2} \frac{\partial \xi_2}{\partial x_2} \quad (18b)$$

Eqs. (18a) and (18b) indicates that the discrete velocity keeps its value and direction unchanged on the physical plane, while varies with the lattice points on the computational plane. The derivatives in Eqs. (18a) and (18b) are calculated by a Jacobi conversion as

$$\begin{bmatrix} \frac{\partial \xi_1}{\partial x_1} & \frac{\partial \xi_1}{\partial x_2} \\ \frac{\partial \xi_2}{\partial x_1} & \frac{\partial \xi_2}{\partial x_2} \end{bmatrix} = \frac{1}{J} \begin{bmatrix} \frac{\partial x_2}{\partial \xi_2} & \frac{\partial x_1}{\partial \xi_2} \\ \frac{\partial x_2}{\partial \xi_1} & \frac{\partial x_1}{\partial \xi_1} \end{bmatrix} \quad (19)$$

where the Jacobian J is determined by

$$J = \frac{\partial x_1}{\partial \xi_1} \frac{\partial x_2}{\partial \xi_2} - \frac{\partial x_1}{\partial \xi_2} \frac{\partial x_2}{\partial \xi_1} \quad (20)$$

For the streaming step, the evolution equations are expressed as

$$g_i(\xi, t + \delta t) = g_i^*(\xi - \Delta \xi_{up,i}, t) \quad (21)$$

$$f_i(\xi, t + \delta t) = f_i^*(\xi - \Delta \xi_{up,i}, t) \quad (22)$$

$$h_i(\xi, t + \delta t) = h_i^*(\xi - \Delta \xi_{up,i}, t) \quad (23)$$

with

$$\Delta \xi_{up,i} = \int_0^{\delta t} \tilde{\mathbf{e}}_i dt \quad (24)$$

where the distance $\Delta \xi_{up,i}$ is calculated by integrating the contravariant velocity over a time step δt , and the two-step Runge-Kutta method is used for solving Eq. (24) as

$$\Delta \xi_{up,i}^{(1)} = \delta t \tilde{\mathbf{e}}_i(\xi)/2, \quad \Delta \xi_{up,i} = \delta t \tilde{\mathbf{e}}_i(\xi - \Delta \xi_{up,i}^{(1)}) + O(\delta t^3) \quad (25)$$

In Eqs. (21)–(23) and (25), the positions $\xi - \Delta \xi_{up,i}$ and $\xi - \Delta \xi_{up,i}^{(1)}$ cannot be directly gained as they are not located at the grid nodes on the computational plane, so an interpolation procedure is extremely needed. In current work, the second-order upwind quadratic interpolation is chosen to guarantee numerical precision, and the interpolation functions are given as

$$g_i(\xi, t + \delta t) = g_i^*(\xi - \Delta \xi_{up,i}, t) = \sum_{m=0}^2 \sum_{l=0}^2 a_{i,m,2} a_{i,l,1} g_i^*(\xi_{i+m-md}, \eta_{j+l-nd}, t + \delta t) \quad (26)$$

$$f_i(\xi, t + \delta t) = f_i^*(\xi - \Delta \xi_{up,i}, t) = \sum_{m=0}^2 \sum_{l=0}^2 a_{i,m,2} a_{i,l,1} f_i^*(\xi_{i+m-md}, \eta_{j+l-nd}, t + \delta t) \quad (27)$$

$$h_i(\xi, t + \delta t) = h_i^*(\xi - \Delta \xi_{up,i}, t) = \sum_{m=0}^2 \sum_{l=0}^2 a_{i,m,2} a_{i,l,1} h_i^*(\xi_{i+m-md}, \eta_{j+l-nd}, t + \delta t) \quad (28)$$

where $md = \text{sgn}(\Delta \xi_{up,i,x})$ and $nd = \text{sgn}(\Delta \xi_{up,i,y})$ are the coefficients to determine the lattice points for interpolation. $a_{i,m,2}$ and $a_{i,l,1}$ are the interpolation coefficients. For the internal lattice points, they are calculated by

$$\begin{aligned} a_{i,0,\alpha} &= (|\Delta \xi_{up,i,\alpha}| - 1)(|\Delta \xi_{up,i,\alpha}| - 2)/2 \\ a_{i,1,\alpha} &= |\Delta \xi_{up,i,\alpha}| (|\Delta \xi_{up,i,\alpha}| - 2) \\ a_{i,2,\alpha} &= |\Delta \xi_{up,i,\alpha}| (|\Delta \xi_{up,i,\alpha}| - 1)/2 \end{aligned} \quad (29)$$

In terms of the neighbor-boundary grid nodes, the interpolation coefficients are determined by

$$\begin{aligned} a_{i,0,\alpha} &= \Delta \xi_{up,i,\alpha} (\Delta \xi_{up,i,\alpha} - 1)/2 \\ a_{i,1,\alpha} &= 1 - \Delta \xi_{up,i,\alpha}^2 \\ a_{i,2,\alpha} &= \Delta \xi_{up,i,\alpha} (\Delta \xi_{up,i,\alpha} + 1)/2 \end{aligned} \quad (30)$$

Finally, the time step δt should be determined by the Courant-Friedrichs-Lewy (CFL) condition as [35]

$$\delta t = \lambda \min_{i,\alpha,\xi} \left| \frac{1}{\tilde{e}_{i,\alpha} \xi} \right| \quad (31)$$

where λ is the CFL number ($\lambda \in (0,1]$). This strategy guarantees that

Table 1
Dimensionless parameters used in the simulation (in lattice units).

Parameter	Value
Density ratio ρ_g/ρ_l	1:1000
Mobility coefficient Γ	450
Surface tension coefficient σ	2.0
Interface width W	2.5
Peclet number Pe	3000

the position of particle after streaming step will not exceed its neighboring lattice cells. Furthermore, basic numerical procedures of present model can refer to the references [36,37] for more details.

3. Validation

The growth of a bubble in overheated liquid without considering gravity is simulated to validate the proposed model in a domain with 300×150 grids. Periodic boundary condition is applied for all the sides. The parameters adopted in simulation are listed in Table 1. Two cases with $Ja = 0.0009$ and $Ja = 0.012$ are calculated, respectively. The obtained bubble equivalent radius at different time steps are compared with the analytical ones of Mikic [38], as shown in Fig. 1. The good agreement demonstrates the reliability of present model in the simulation of problems accompanied with liquid-gas phase transition.

4. Numerical results and discussion

4.1. Simulation set up

In this section, the motion characteristic of a single bubble rising along different overheated surfaces will be investigated to explore the impact of surface configuration on heat and mass transfer. Initially, an elliptic cylinder obstacle of axial ratio a/b is immersed in a cavity full of liquid with saturated temperature T_s . And, half of the domain is considered in present work as shown in Fig. 2. A bubble with a diameter of 0.15 is located at the point (0.2, 1.16), whose inner temperature is assumed to be T_s as well. At a specific time, the cylinder surface is heated to a high temperature T_w ($T_w > T_s$). Consequently, phase transition occurs as the bubble rises to the vicinity of cylinder surface. The boundary settings are labelled in Fig. 2, and all the boundary conditions can be implemented by the non-equilibrium extrapolation

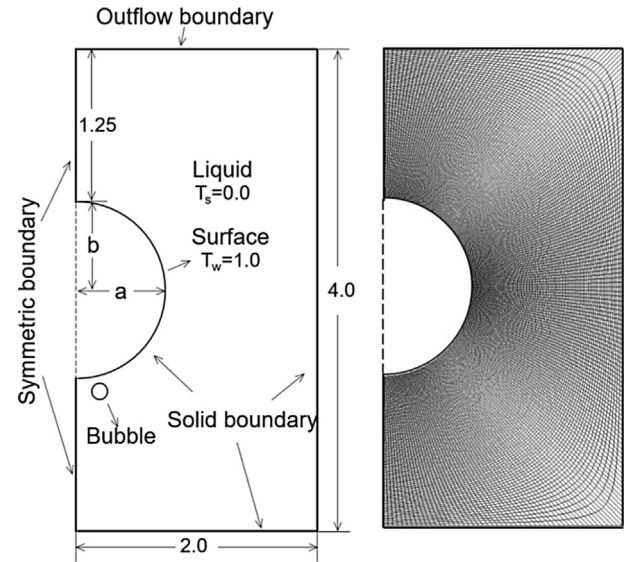


Fig. 2. Illustration for research domain (left) and grid generation (right).

scheme [39,40].

Before the simulation is carried out, the effect of grid resolution should be studied. Research indicates that the deformation of bubble is significantly influenced by the width of computational domain, and such influence could be neglected if the width of domain is larger than four times of the bubble diameter [34,41]. Therefore, the lattice system of 400×800 in body-fitted coordinates is adopted in current study. The key parameters used in simulation are all listed in Table 2. Three cases are considered for the numerical simulations, and the respective surface axial ratios a/b are 0.5, 1.0, and 1.5.

During the sliding and rising process, the velocity of bubble is not a constant. Its spatial average horizontal component V_h and vertical component V_v are evaluated by [22]

$$V_h = \frac{\sum v_x \phi}{\sum \phi}, \quad V_v = \frac{\sum v_y \phi}{\sum \phi}, \quad \text{where } \phi < 0 \quad (32)$$

where v_x and v_y are the local horizontal and vertical components of gas velocity, respectively. Furthermore, in order to avoid the effect of initial spontaneous diffusion of the bubble, the temperature field and buoyancy are imposed after a short time (e.g., 2000 time steps).

4.2. Results and analysis

In the present work, the buoyancy acted on the bubble is tuned by the Eötvös number Eo as

$$Eo = \frac{g(\rho_l - \rho_g)D_0^2}{\sigma} \quad (33)$$

Fig. 3 shows the evolution of bubble shape in its rising and sliding

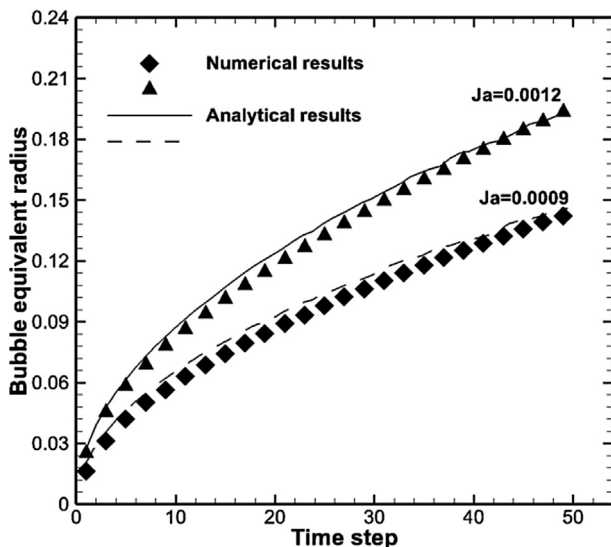


Fig. 1. Comparison of the bubble equivalent radius with the analytical results.

Table 2

Key parameters adopted in current simulation (in lattice units).

Parameter	Value
Density ratio ρ_g/ρ_l	1:1000
Mobility coefficient Γ	500
Surface tension coefficient σ	15
Interface width W	5.5
Initial diameter of bubble D_0	0.15
Surface temperature T_w	1.0
Saturated temperature of liquid and bubble T_s	0.0
Jakob number Ja	0.0003
Peclet number Pe	3000
Eötvös number Eo	0.33

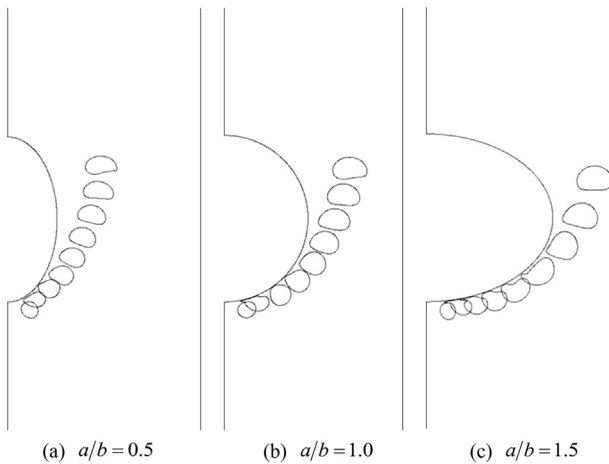


Fig. 3. Evolution of bubble shape in the rising and sliding process on overheated curved surfaces with various axial ratios ($\Delta t = 12,000$).

process along overheated curved surfaces with various axial ratios. The time interval between two adjacent bubbles is $\Delta t = 12,000$ time steps. By comparing the three evolution processes in Fig. 3, the existence of cylinder obstacle results in large deformation of bubble shape, and the deformation increases with the surface axial ratio. Meanwhile, since the near-wall liquid absorbs heat from the hot surface and evaporates, the bubble grows consequently. Along with the upward movement of the growing bubble, the elongated tail part of the bubble caused by surface vanishes gradually. At a specified time, the bubble departs from the surface and rises toward the bulk flow. Additionally, for the case with bigger surface axial ratio, the time of bubble sliding on the surface is longer, which results in larger growth of bubble than other cases.

Figs. 4 and 5 show the variations of bubble average horizontal and vertical velocities with time step, respectively. Initially, under the effects of buoyancy, surface tension, and viscous force, both the average horizontal and vertical velocities increase before the rising bubble contacts with the surface. Once the bubble slides along the surface, its average horizontal velocity continues to increase under the effect of dominated lateral force from the surface. In terms of the average vertical velocity, it declines due to the vertical resistance from the surface at the beginning, and then increases because the surface effect is weakened as the bubble rising upward. When the bubble departs from the surface, the average horizontal velocity begins to decrease, while the

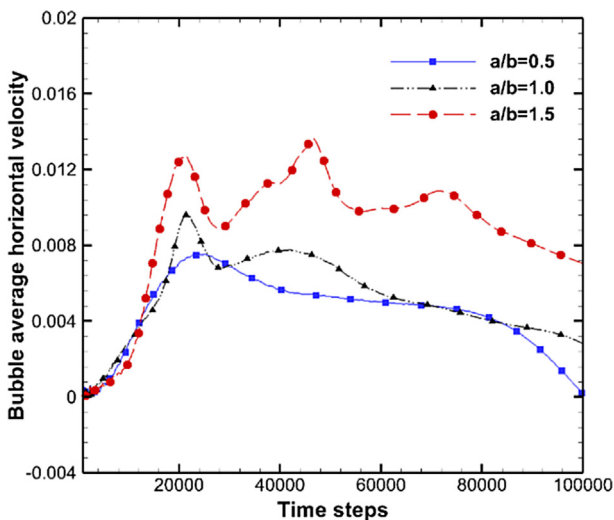


Fig. 4. Time variation of bubble average horizontal velocity for cases with various surface axial ratios.

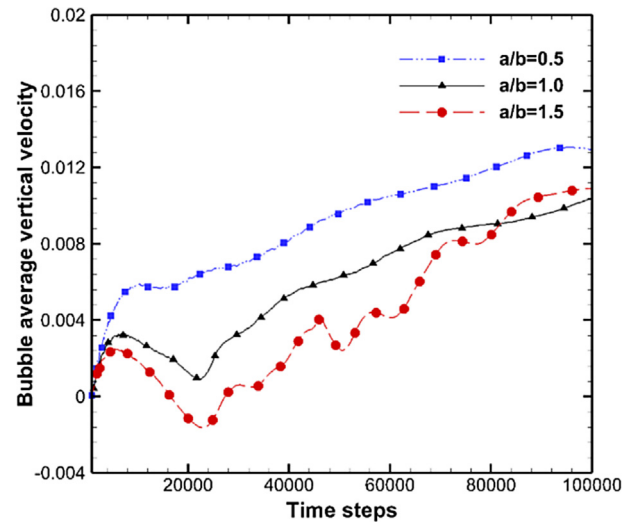


Fig. 5. Time variation of bubble average vertical velocity for cases with various surface axial ratios.

increasing tendency is kept for the average vertical one. Moreover, fluctuation is observed in both variation curves of horizontal and vertical average velocities, which stems from the deformation and expansion of the bubble. For different cases, the horizontal component of bubble average velocity increases with the surface axial ratio, yet opposite for the vertical one. The reason is that the lateral force which accelerates average horizontal velocity is bigger from the surface with a larger axial ratio. Meanwhile, the corresponding vertical resistance increases with surface axial ratio, which suppresses the rising tendency of average vertical velocity.

The evolutions of temperature field for the three cases with different surface axial ratios are presented in Fig. 6. It is noted that the bubble motion at different stages has significant effects on the temperature field. Initially, transient conduction from the hot surface to the liquid forms an overheated layer on the surface. During the bubble sliding stage, contact area between the bubble and the thin liquid film upon the surface is the largest. Micro-evaporation occurs underneath the bubble across the thin liquid film [13]. Combined with macro-evaporation of the liquid immersed in the overheated layer, the bubble grows rapidly and its influence on temperature becomes obviously. As presented in both Figs. 3 and 6, the time of bubble trapped in overheated layer is longer when it slides on the surface with a larger axial ratio. The consequent larger deformation and growth of bubble lead to more drastic changing of temperature field than other cases.

In order to quantitatively analyze the effect of bubble evolution on temperature field, the local Nusselt number (Nu) distribution around overheated surface is calculated by

$$Nu = -\frac{L}{T_w - T_s} \left(\frac{\partial T}{\partial n} \right)_{\text{surface}} \quad (34)$$

where L is the characteristic length of the surface, $(\partial T / \partial n)_{\text{surface}}$ is the temperature gradient on the surface. For the similarity, the local Nusselt number distribution around overheated surface with $a/b = 1.0$ is chosen to present as example. A comparison of four typical distributions corresponding to different bubble positions is made in Fig. 7. It can be found out that when the bubble contacts with the surface during its sliding stage, a peak in local Nusselt number distribution curve is observed, and the peak location is related to the bubble position. The reason is that the existence of bubble causes an extremely high temperature gradient on the surface as the inner temperature of bubble maintains to be relatively low and stable. When the bubble departs from the surface, the disturbance of near-wall temperature field aroused by bubble is weakened and the peak value of local Nusselt number

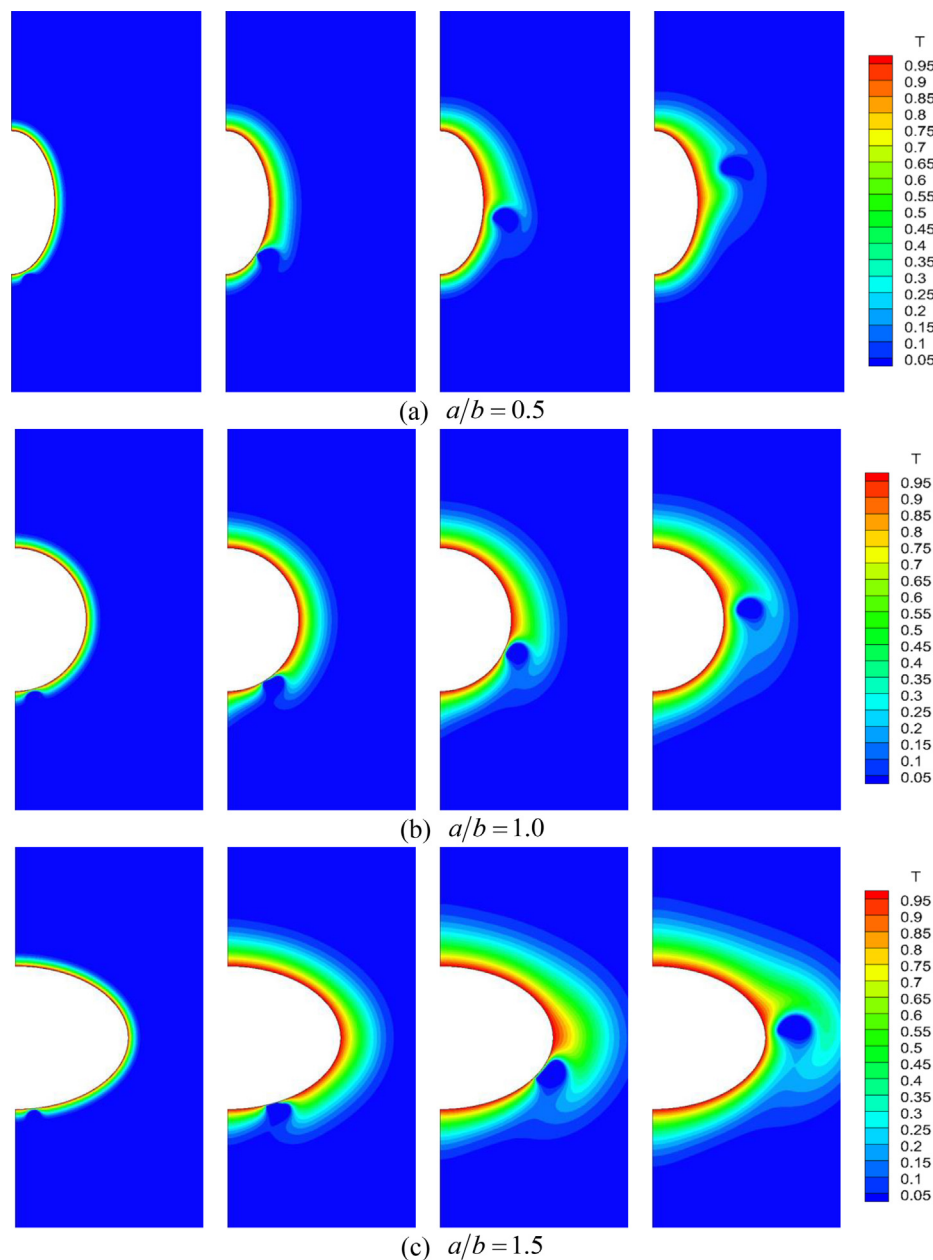


Fig. 6. Evolution of temperature field for cases with various surface axial ratios (time step: $6000 \leq t \leq 96,000$, $\Delta t = 24,000$).

begins to decline. Eventually, the local Nusselt numbers recover to be evenly distributed on the surface when the bubble moves to the far field (Position 4 in Fig. 7b).

5. Conclusions

In the present work, a two-phase thermal LBM model for simulating liquid-gas phase transition in body-fitted coordinates is proposed by combining the hybrid thermal LBM model in Cartesian coordinates with the GILBM scheme. In order to verify the ability of present model in simulating liquid-gas phase transition, the growth of bubble in overheated liquid without considering gravity is calculated. The results are in good agreement with the theoretical solutions of Mikic [38].

After that, the model is applied to study the motion characteristic of a single bubble rising along overheated surfaces with different curvatures. Three kinds of elliptic surfaces with different axial ratios have been investigated to explore its impact on heat and mass transfer. During the rising and sliding process, the bubble deforms mainly due to

the existence of the curved surface, while it grows as the micro-evaporation across the thin liquid film and the macro-evaporation of the liquid immersed in the overheated layer. In addition, the bubble experiences the largest deformation and growth in the case with surface axial ratio $a/b = 1.5$. As for the variation of bubble average velocity, both the horizontal and vertical components increase initially. But, at the stage of bubble sliding along the surface, the horizontal component continues to increase while the vertical one decreases. The variation tendency converts after the bubble departs from the surface. For different cases, the horizontal component increases with the surface axial ratio, while the vertical one is opposite. The influences of bubble deformation and expansion on temperature field become more significant when bubble rising along surface with a larger axial ratio. Simultaneously, the existence of bubble causes an extremely high temperature gradient on the surface which breaks the evenness of local Nusselt number distribution. When the bubble departs from the surface to the far field, the disturbance of near-wall temperature field aroused by bubble is weakened and the local Nusselt numbers around surface

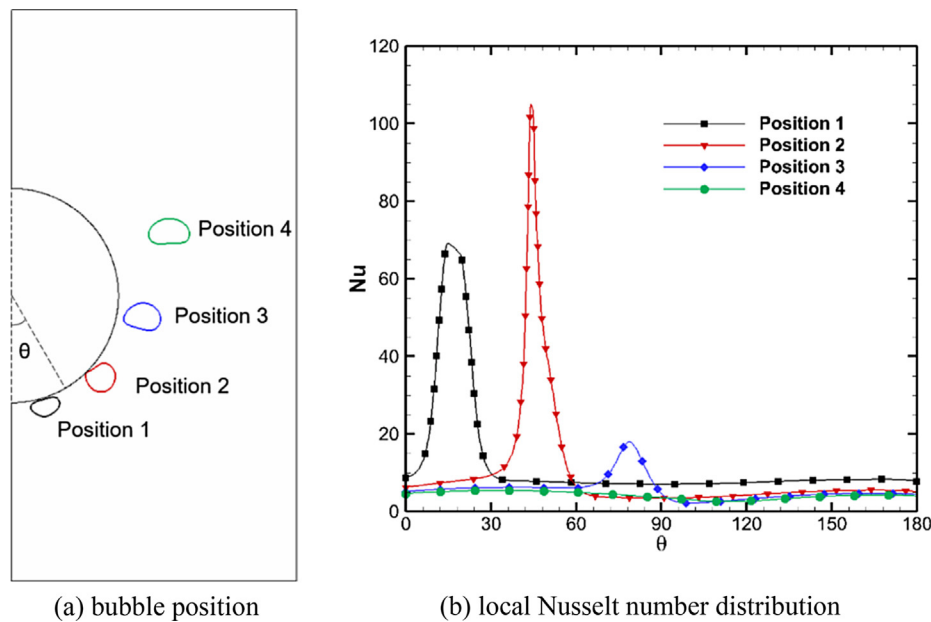


Fig. 7. Local Nusselt number distributions around overheated surface with $a/b = 1.0$ for various bubble positions.

recover to be evenly distributed. The successful application of present model in above investigation suggests that it is reliable to use the present model to study liquid-gas phase transition in curved/complicated domains, which bridges the gap between the LBM and practical applications further.

6. Conflict of interest statement

None.

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